

Victor Obarafor, PhD

Researcher in Trustworthy AI | Applied AI Scientist | Lecturer in Computing
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Professional Summary

AI researcher specializing in privacy-preserving machine learning, federated systems, and AI security. Four years of dual academic and applied research experience developing scalable frameworks for secure distributed intelligence in smart environments. Proven record of peer-reviewed IEEE publications, cross-disciplinary collaboration, and guiding advanced research to deployment-ready outcomes. Seeking roles in applied AI research or trustworthy ML to translate foundational privacy research into impactful industry solutions.

Experience

Canterbury Christ Church University, UK

Artificial Intelligence Researcher | Nov 2021 – Present

- Designed and implemented privacy-preserving frameworks for federated learning and IoT, advancing data protection in large-scale distributed AI.
- Developed ConfidenShuffle, a novel architecture reducing gradient leakage and enhancing privacy robustness; published in IEEE Access (2025).
- Pioneered adversarial-resilient AI pipelines for real-world IoT deployments, improving model robustness against inference attacks.
- Led multi-disciplinary teams bridging AI, cybersecurity, and systems engineering to produce deployable privacy solutions adopted in smart environment testbeds.
- Published 3 first-author papers in IEEE Access (2023–2025); research cited in international projects on secure intelligent systems.

Lecturer in Computing | Nov 2021 – Present

- Delivered and led 12+ modules across cybersecurity, AI, and computing, integrating real-time cloud deployments (AWS, Security+, Docker, PyTorch).
- Supervised advanced student research in federated learning, adversarial ML, and IoT security, producing deployable prototype systems.
- Directed curriculum modernization toward AI ethics, trustworthy systems, and secure edge intelligence, aligning with industry standards.
- Mentored undergraduate and postgraduate researchers into first-author publications and academic conferences.

Education

PhD, Artificial Intelligence (Federated Learning for Smart Environments)

Canterbury Christ Church University, UK | 2021 – 2024

MSc, Cyber Security and Forensics (Distinction)

University of Westminster, UK | 2020 – 2021

BSc, Information Technology (First Class Honours)

University of Ghana | 2012 – 2016

Publications (IEEE Access)

- Advancing IoT for Smart Environments Using the ConfidenShuffle Framework* (2025)
- Privacy-Preserving Techniques for IoT Systems Using Federated Learning* (2023)
- Feedforward ANN-Based Intrusion Detection System for Secure IoT Applications* (2025)

Technical Expertise

AI/ML: PyTorch, TensorFlow, Scikit-learn, Federated Learning, Adversarial ML, Deep Neural Networks

Security: Differential Privacy, Homomorphic Encryption, Intrusion Detection, Privacy-Enhancing Tech

Cloud/DevOps: AWS, Docker, Kubernetes, Git, CI/CD, API Integration

Research Tools: Python, NumPy, Pandas, MATLAB, Jupyter, LaTeX, IEEE Xplore Standards

References

Available on Request